

Monte Carlo Coulomb Collisions for Guiding-Centre Particles in Boozer Coordinates: Implementation Notes

firm3d — ep_thermal_collisions

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1 Overview

These notes describe the algorithm used in `trace_particles_boozer_with_collisions` to include stochastic Coulomb collisions for an energetic particle (EP) tracing through one or more Maxwellian background species.

The approach couples the deterministic guiding-centre (GC) orbit equations in Boozer coordinates with a Milstein stochastic term for the velocity-space diffusion. The full state vector is $(s, \theta, \zeta, v, \xi)$, where $v = |\mathbf{v}|$ is the total speed and $\xi = v_{\parallel}/v$ is the pitch.

Primary references:

- Hirvijoki et al. [1] — Monte Carlo implementation of the guiding-centre Fokker–Planck equation in (v, ξ) coordinates; Eqs. (32)–(33) of that paper define the SDE system used here.
- Hirvijoki et al. [2] — ASCOT4/5 code description; source file `mccc_coefs.h` defines the Coulomb logarithm convention adopted here.
- Boozer & Kuo-Petravic [4] — pitch-angle representation in Boozer coordinates.

2 Guiding-Centre Equations in Boozer Coordinates

In Boozer coordinates (s, θ, ζ) , where s is the normalised toroidal flux, θ the poloidal angle, and ζ the toroidal angle, the vacuum GC equations are

$$\dot{s} = -\frac{m v^2 (1 + \xi^2)}{2 q \psi_0} \frac{\partial B}{\partial \theta} \frac{1}{B}, \quad (1)$$

$$\dot{\theta} = \frac{m v^2 (1 + \xi^2)}{2 q \psi_0} \frac{\partial B}{\partial s} \frac{1}{B} + \frac{\iota \xi v B}{G}, \quad (2)$$

$$\dot{\zeta} = \frac{\xi v B}{G}, \quad (3)$$

$$\dot{\xi} = -\frac{(\iota \partial_{\theta} B + \partial_{\zeta} B) v (1 - \xi^2)}{2G} \quad [\text{mirror force}], \quad (4)$$

where ψ_0 is the reference poloidal flux, $G(s)$ is the covariant toroidal field component, and $\iota(s)$ is the rotational transform. The magnetic moment $\mu = v^2(1 - \xi^2)/(2B)$ is reconstructed locally at each evaluation.

With collisions the equations for v and ξ acquire deterministic drift terms (Section 4) and stochastic noise terms (Section 5.2).

3 Background Species and Profiles

Each background species b is described by a `ThermalBackground` object specifying:

- $n_b(s)$ — number density [m^{-3}], callable of the flux label $s \in [0, 1]$;
- $T_b(s)$ — temperature [J], callable (use `T_keV × 1e3 × e`);
- m_b — species mass [kg];
- q_b — species charge [C] (signed).

Internally the profiles are pre-evaluated on a uniform s -grid of 512 points and linearly interpolated in C++.

4 Collision Coefficients

Notation

The symbols used here follow the ASCOT5 source code (`mccc_coefs.h`, [3]); Table 1 shows the correspondence with Hirvijoki et al. [1], whose Eqs. (32)–(33) define the SDE structure.

This paper	ASCOT5 <code>mccc_coefs.h</code>	Hirvijoki et al. [1]
$D_{\parallel}$	<code>Dpara</code>	$D_k/2$
ν_D	<code>nu</code> = $2D_{\perp}/v^2$	$2D_{\perp}/v^2$
Q	<code>Q</code>	(Einstein-relation drag in K)
K	<code>K</code> = $Q + D'_{\parallel} + 2D_{\parallel}/v$	K (drift in Eq. 32)

Table 1: Notation correspondence. The noise amplitude for v is $g_v = \sqrt{2D_{\parallel}}$ (this paper) = $\sqrt{D_k}$ (Hirvijoki 2013).

4.1 Debye Length

The total Debye screening length from all background species is [5, 6]

$$\frac{1}{\lambda_D^2} = \sum_b \frac{n_b q_b^2}{\varepsilon_0 T_b}. \quad (5)$$

4.2 Coulomb Logarithm

The Coulomb logarithm is defined as the ratio of maximum to minimum impact parameters [5, 6, 7]:

$$\ln \Lambda_{ab} = \ln \left(\frac{\lambda_D}{b_{\min}} \right), \quad b_{\min} = \max(b_{\text{cl}}, b_{\text{qm}}). \quad (6)$$

The maximum impact parameter is the Debye length (Eq. (5)). The minimum impact parameter is the larger of the classical 90°-deflection distance and the quantum-mechanical de Broglie

wavelength of the reduced mass:

$$b_{\text{cl}} = \frac{|q_a q_b|}{4\pi\epsilon_0 m_r \bar{v}^2}, \quad (7)$$

$$b_{\text{qm}} = \frac{\hbar}{2m_r \sqrt{\bar{v}^2}}, \quad (8)$$

where $m_r = m_a m_b / (m_a + m_b)$ is the reduced mass.

The effective velocity scale $\bar{v}^2$ is taken as

$$\bar{v}^2 = v^2 + v_{\text{th},b}^2, \quad v_{\text{th},b} = \sqrt{2T_b/m_b}, \quad (9)$$

following the ASCOT5 source convention [3]. This form is a numerical interpolation device, not derived from first principles: Ichimaru [7] derives b_{cl} and b_{qm} only in the fast-particle limit $v \gg v_{\text{th},b}$ (Section 4.3B of that reference), where $\bar{v} \approx v$. The $\bar{v}^2$ form extends the formula to the slow-particle regime without a discontinuous switch: fast EP against ions ($v \gg v_{\text{th},b}$, $\bar{v} \approx v$) and slow EP against electrons ($v \ll v_{\text{th},e}$, $\bar{v} \approx v_{\text{th},e}$).

The quantum branch $b_{\text{qm}} > b_{\text{cl}}$ applies when

$$\frac{2\pi\epsilon_0 \hbar \sqrt{\bar{v}^2}}{|q_a q_b|} > 1, \quad (10)$$

i.e. when the de Broglie wavelength of the reduced mass particle exceeds the classical distance of closest approach. For 3.52 MeV α -particles scattering off 10 keV electrons one finds $b_{\text{qm}} \approx 22 b_{\text{cl}}$, so the quantum branch is active throughout alpha slowing-down against electrons. If $\ln \Lambda \leq 0$ (Debye length below the minimum impact parameter, e.g. $T_b \rightarrow 0$ at finite density) the binary-collision model is undefined and an exception is raised; ASCOT5 reaches the equivalent outcome by aborting markers through its input-evaluation errors, and its input profiles keep the temperature finite wherever the density is nonzero. A warning is printed for $0 < \ln \Lambda < 2$, where the perturbative expansion underlying the Fokker–Planck operator begins to lose validity.

4.3 Chandrasekhar G Function

The Chandrasekhar function [8] is

$$G(x) = \frac{\text{erf}(x) - \frac{2x}{\sqrt{\pi}} e^{-x^2}}{2x^2}, \quad x = v/v_{\text{th},b}. \quad (11)$$

Its derivative satisfies the recurrence

$$G'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2} - \frac{2G(x)}{x}. \quad (12)$$

Key limiting behaviours:

$$\begin{aligned} x \rightarrow 0 : \quad & G \rightarrow 0, \quad G' \rightarrow 2/(3\sqrt{\pi}) \approx 0.376; \\ x \rightarrow \infty : \quad & G \rightarrow 1/(2x^2), \quad G' \rightarrow 0. \end{aligned}$$

4.4 Collision Frequency Prefactor

Define the Rosenbluth prefactor for species b [9, 10]:

$$\Gamma_b = \frac{n_b q_a^2 q_b^2 \ln \Lambda_{ab}}{4\pi\epsilon_0^2 m_a^2}. \quad (13)$$

4.5 Pitch-Angle Scattering Frequency

$$\nu_D^{(b)} = \frac{\Gamma_b [\operatorname{erf}(x) - G(x)]}{v^3} \equiv \frac{2D_\perp^{(b)}}{v^2}. \quad (14)$$

The second form uses the perpendicular diffusion coefficient $D_\perp^{(b)} = \Gamma_b [\operatorname{erf}(x) - G(x)]/(2v)$ and matches ASCOT5's `mccc_coefs_nu` and the $2D_\perp/v^2$ notation of Hirvijoki et al. [1, 2].

4.6 Parallel Diffusion Coefficient

$$D_\parallel^{(b)} = \frac{\Gamma_b G(x)}{v}, \quad \frac{\partial D_\parallel^{(b)}}{\partial v} = \frac{\Gamma_b}{v} \left(\frac{G'(x)}{v_{\text{th},b}} - \frac{G(x)}{v} \right). \quad (15)$$

In Hirvijoki et al. [1] the parallel diffusion coefficient is denoted D_k , with $D_\parallel = D_k/2$; the noise amplitude for the speed SDE is $\sqrt{D_k} = \sqrt{2D_\parallel}$ in both conventions. ASCOT5 names this coefficient `dpara`. The v -derivative is required by the Milstein correction term (Section 5.2).

4.7 Deterministic Drag in v

The drag coefficient (ASCOT5: `mccc_coefs_Q` [3]) is fixed by the Einstein relation:

$$Q^{(b)} = -\frac{m_a v}{T_b} D_\parallel^{(b)} = -\Gamma_b \frac{m_a G(x)}{T_b}. \quad (16)$$

The total deterministic drift in speed is

$$K^{(b)} = Q^{(b)} + \frac{\partial D_\parallel^{(b)}}{\partial v} + \frac{2D_\parallel^{(b)}}{v} \quad (17)$$

(ASCOT5's `mccc_coefs_K` [3]; the drift coefficient K of Eq. (32) in Hirvijoki et al. [1]).

Itô transformation leading to Eq. (17)

Equation (17) can be obtained two independent ways, which cross-check each other.

(i) **Itô's formula on $v = |\mathbf{v}|$.** In three-dimensional velocity space the collisional motion of the test particle is

$$d\mathbf{v} = F_{\text{dyn}} \hat{\mathbf{v}} dt + \boldsymbol{\sigma} \cdot d\mathbf{W}, \quad \boldsymbol{\sigma} \boldsymbol{\sigma}^\top = 2D_\parallel \hat{\mathbf{v}} \hat{\mathbf{v}} + 2D_\perp (\mathbf{I} - \hat{\mathbf{v}} \hat{\mathbf{v}}), \quad (18)$$

where F_{dyn} is the dynamical friction [8, 10, 11]. The speed $v = |\mathbf{v}|$ is a nonlinear function of $\mathbf{v}$, so Itô's formula applies with $\nabla_v v = \hat{\mathbf{v}}$ and Hessian $\nabla_v \nabla_v v = (\mathbf{I} - \hat{\mathbf{v}} \hat{\mathbf{v}})/v$:

$$dv = \left(F_{\text{dyn}} + \frac{2D_\perp}{v} \right) dt + \sqrt{2D_\parallel} dW_v. \quad (19)$$

The $2D_\perp/v$ drift is the Bessel-process effect: kicks transverse to $\mathbf{v}$ do not change v at first order, but they lengthen the vector at second order, $\Delta v = |\mathbf{v} + \Delta \mathbf{v}_\perp| - v \approx |\Delta \mathbf{v}_\perp|^2/2v > 0$.

(ii) Fokker–Planck matching (detailed balance). The Itô SDE $dv = K dt + \sqrt{2D_{\parallel}} dW_v$ evolves the speed density $P(v) \propto v^2 f(v)$ according to $\partial_t P = -\partial_v(KP) + \partial_v^2(D_{\parallel}P)$. Zero flux on the equilibrium $P_{\text{eq}} \propto v^2 e^{-m_a v^2/2T_b}$ requires $KP_{\text{eq}} = \partial_v(D_{\parallel}P_{\text{eq}})$, i.e.

$$K = -\frac{m_a v}{T_b} D_{\parallel} + \frac{\partial D_{\parallel}}{\partial v} + \frac{2 D_{\parallel}}{v}, \quad (20)$$

which is Eq. (17) with Q as in Eq. (16). Here $\partial_v D_{\parallel}$ is the multiplicative-noise (Itô) drift that appears when the diffusion term is commuted into advective form, and $2D_{\parallel}/v$ comes from the Jacobian v^2 of spherical velocity-space coordinates.

Consistency of (i) and (ii). For Maxwellian field particles the dynamical friction obeys the identity

$$F_{\text{dyn}}^{(b)} = -\left(1 + \frac{m_a}{m_b}\right) \frac{2x^2 G(x) \Gamma_b}{v^2} = Q^{(b)} + \frac{\partial D_{\parallel}^{(b)}}{\partial v} + \frac{2(D_{\parallel}^{(b)} - D_{\perp}^{(b)})}{v}, \quad (21)$$

so that $F_{\text{dyn}} + 2D_{\perp}/v = Q + \partial_v D_{\parallel} + 2D_{\parallel}/v = K$ exactly. (F_{dyn} is ASCOT5's `mccc_coefs_F`, which is not used in the guiding-centre scheme.) In the fast-particle limit $x \gg 1$ this identity encodes the classic cancellation

$$F_{\text{dyn}} + \frac{2D_{\perp}}{v} \longrightarrow -\left(1 + \frac{m_a}{m_b}\right) \frac{\Gamma_b}{v^2} + \frac{\Gamma_b}{v^2} = -\frac{m_a}{m_b} \frac{\Gamma_b}{v^2} :$$

most of the dynamical friction is compensated by transverse diffusion, and the net slowing-down rate is set by the mass ratio. Note in particular that Q is *not* the dynamical friction — it is the Einstein-relation drag $-(m_a v/T_b)D_{\parallel}$.

4.8 Summation over Species

All coefficients are additive over background species:

$$\nu_D = \sum_b \nu_D^{(b)}, \quad D_{\parallel} = \sum_b D_{\parallel}^{(b)}, \quad K = \sum_b K^{(b)}. \quad (22)$$

The Debye length used in Eq. (6) is computed from all species simultaneously (Eq. (5)), prior to the per-species loop.

5 Stochastic Differential Equations

5.1 Full SDE System

The complete SDE for the five-dimensional state $(s, \theta, \zeta, v, \xi)$ is [1]

$$ds = \dot{s}_{\text{orb}} dt, \quad (23)$$

$$d\theta = \dot{\theta}_{\text{orb}} dt, \quad (24)$$

$$d\zeta = \dot{\zeta}_{\text{orb}} dt, \quad (25)$$

$$dv = K(v, s) dt + \sqrt{2D_{\parallel}} dW_v, \quad (26)$$

$$d\xi = \dot{\xi}_{\text{mirror}} dt - \xi \nu_D dt + \sqrt{\nu_D(1 - \xi^2)} dW_{\xi}, \quad (27)$$

where dW_v and dW_ξ are independent Wiener increments with $\mathbb{E}[dW^2] = dt$. Equations (26)–(27) correspond to Eqs. (32)–(33) of Hirvijoki et al. [1], with the identification $D_\parallel \leftrightarrow D_k/2$ and $\nu_D \leftrightarrow 2D_\perp/v^2$ in their notation. The noise amplitude for ξ ensures that the marginal distribution of ξ converges to a uniform distribution on $[-1, 1]$ in the isotropisation limit $\nu_D t \gg 1$.

5.2 Milstein Scheme

After each accepted Dormand–Prince (DP) step of physical size h , the stochastic increments $\Delta W_v \sim \mathcal{N}(0, h)$ and $\Delta W_\xi \sim \mathcal{N}(0, h)$ are generated and applied via the Milstein method [12]:

$$v \leftarrow v + g_v \Delta W_v + \frac{1}{2} \frac{\partial g_v}{\partial v} (\Delta W_v^2 - h), \quad (28)$$

$$\xi \leftarrow \xi + g_\xi \Delta W_\xi + \frac{1}{2} \frac{\partial g_\xi}{\partial \xi} (\Delta W_\xi^2 - h), \quad (29)$$

where the noise amplitudes and their derivatives are

$$g_v = \sqrt{2D_\parallel}, \quad \frac{\partial g_v}{\partial v} = \frac{\partial D_\parallel / \partial v}{\sqrt{2D_\parallel}}, \quad (30)$$

$$g_\xi = \sqrt{\nu_D(1 - \xi^2)}, \quad \frac{\partial g_\xi}{\partial \xi} = \frac{-\xi \nu_D}{\sqrt{\nu_D(1 - \xi^2)}}. \quad (31)$$

In terms of the coefficient structs, the Milstein correction for ξ simplifies to

$$\delta \xi_{\text{Milstein}} = -\frac{1}{2} \xi \nu_D (\Delta W_\xi^2 - h). \quad (32)$$

Boundary conditions follow ASCOT5 (`mccc_gc_milstein.c`, [3]). The speed reflects off a thermal cutoff,

$$v < v_{\text{cut}} \Rightarrow v \leftarrow 2v_{\text{cut}} - v, \quad v_{\text{cut}} = 0.1 \sqrt{T_{\text{min}}/m_a}, \quad (33)$$

where T_{min} is the coldest active background species (ASCOT5’s `MCCC_CUTOFF` uses its first background species), so that the collision coefficients, which diverge as $v \rightarrow 0$, are never evaluated deep below the thermal speed. The pitch mirrors at the boundary: $|\xi| > 1 \Rightarrow \xi \leftarrow \text{sign}(\xi) (2 - |\xi|)$; a hard clamp would pile up probability at exactly $\xi = \pm 1$.

6 Adaptive Integration Strategy

6.1 Dormand–Prince Solver

The deterministic ODE part (all five components of the drift) is integrated with the classic 4(5)-order embedded Dormand–Prince (FSAL) method [13]. Step-size control uses mixed absolute/relative tolerances:

$$\text{err}_i = |y_i^{(5)} - y_i^{(4)}| \leq \delta_{\text{abs}} + \delta_{\text{rel}} |y_i|. \quad (34)$$

Default tolerances: $\delta_{\text{abs}} = \delta_{\text{rel}} = 10^{-9}$.

6.2 Minimum Step Size

The parameter `DP_hmin` (default 0.0) sets a lower bound on the physical step size h in seconds. A value of 10^{-10} – 10^{-9} s prevents the solver from stalling at bounce points while keeping steps small enough that the orbital dynamics remain accurate.

6.3 Operator Splitting

The SDE is solved via Lie–Trotter operator splitting:

1. **Deterministic half:** advance $(s, \theta, \zeta, v, \xi)$ by one DP step of physical time h using the full drift $(\dot{s}, \dot{\theta}, \dot{\zeta}, K, -\xi\nu_D + \dot{\xi}_{\text{mirror}})$.
2. **Stochastic half:** apply Milstein noise to (v, ξ) at the accepted state, using h as the diffusion window.

The trajectory is saved *before* the Milstein step at each t_{save} checkpoint, so saved states correspond to the deterministic prediction (consistent with the standard output convention).

7 Output Format

Each particle trace is a NumPy array of shape $(\text{nimesteps}, 6)$ with columns

$$[t, s, \theta, \zeta, v_{\parallel}, v].$$

From these the kinetic energy $E = \frac{1}{2}mv^2$ and the magnetic moment $\mu = (v^2 - v_{\parallel}^2)/(2B)$ can be reconstructed at each saved point. When `forget_exact_path=True` only the first and last rows are retained.

Each particle uses RNG seed `rng_seed+i`, making multi-particle MPI runs fully reproducible.

8 Python Interface

```
from firm3d.field.collisions import ThermalBackground, \
    trace_particles_boozers_with_collisions
from firm3d.util.constants import (
    PROTON_MASS, ELECTRON_MASS, ELEMENTARY_CHARGE, ONE_EV,
    ALPHA_PARTICLE_MASS, ALPHA_PARTICLE_CHARGE,
    FUSION_ALPHA_PARTICLE_ENERGY,
)

# Typical DT-fusion plasma backgrounds
n0 = 1e20          # m^-3
deuterium = ThermalBackground(
    n_profile=lambda s: n0 * (1 - s),
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=2 * PROTON_MASS,
    charge=ELEMENTARY_CHARGE,
)
tritium = ThermalBackground(
    n_profile=lambda s: n0 * (1 - s),
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=3 * PROTON_MASS,
    charge=ELEMENTARY_CHARGE,
)
```

```

electrons = ThermalBackground(
    n_profile=lambda s: 2 * n0 * (1 - s), # quasi-neutral
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=ELECTRON_MASS,
    charge=-ELEMENTARY_CHARGE,
)

res_tys, res_hits = trace_particles_boozers_with_collisions(
    field,
    stz_inits, # (N, 3) array
    parallel_speeds, # (N,) array, m/s
    backgrounds=[deuterium, tritium, electrons],
    tmax=1e-1, # 100 ms
    mass=ALPHA_PARTICLE_MASS,
    charge=ALPHA_PARTICLE_CHARGE,
    Ekin=FUSION_ALPHA_PARTICLE_ENERGY, # 3.52 MeV
    tol=1e-9,
    dt_save=1e-4, # save every 0.1 ms
    DP_hmin=1e-10, # s
    rng_seed=42,
)

```

9 Testing and Validation

The implementation is exercised at four levels of increasing scope: analytic unit tests of the coefficient formulae, pure-Python velocity-space relaxation tests with no compiled module or magnetic field, end-to-end tests of the full production tracer (guiding-centre orbits + Milstein noise), and cross-code validation against ASCOT5 [3] in both a controlled uniform plasma and a full reactor-scale 3D stellarator equilibrium. All numbers below are reproducible from `tests/field/test_collisions.py`, `tests/field/test_collisions_local.py`, and the examples referenced in each subsection.

9.1 Analytic Coefficient Unit Tests

`TestCollisionCoefficients` checks the Chandrasekhar function $G(x)$ and its derivative against known limits ($G(0) = 0$, $G(x \rightarrow \infty) \rightarrow 1/2x^2$, sign change of G' near $x \approx 0.92$), verifies $K < 0$ (drag always decelerates) and $\nu_D > 0$ for arbitrary finite backgrounds, and checks the expected scalings $K \propto n_b$, $K \propto q_a^2$, $\nu_D \propto n_b$, and additivity across species to $\lesssim 10\%$ (the residual comes from the $\ln \Lambda$ dependence on λ_D). `TestUnphysicalCoulombLog` confirms that $\ln \Lambda \leq 0$ raises an exception rather than returning silently unphysical coefficients (Section 4.2). These are deterministic, sub-second checks with no random component.

9.2 Local Velocity-Space Relaxation Tests

`tests/field/test_collisions_local.py` mirrors `collisions.h` in pure Python (no compiled module, no magnetic field) and integrates only the velocity-space SDE (Eqs. (26)–(27) with the orbital drifts set to zero) for an ensemble of $N = 4000$ particles. All collision rates scale linearly

with the background density, so boosting $n_b \rightarrow 10^{23} \text{ m}^{-3}$ compresses the relaxation time to microseconds, letting the full suite run in about 10 seconds on a laptop. These tests are the most direct regression check on the Einstein-relation drag (Eq. (16)): an earlier, incorrect form of Q passed all analytic unit tests but drove a proton ensemble to $\langle E \rangle = 3.7 T_b$ instead of $1.5 T_b$ here, with a Kolmogorov–Smirnov (KS) p -value of 0 against the Maxwell speed law.

Test	$\langle E \rangle / T_b$ (target 1.5)	KS p -value	Pass
Cold start ($E_0 = 0.45 T_b$)	1.446	0.037	✓
Hot start ($E_0 = 4.5 T_b$)	1.486	0.095	✓
Stationarity (start at equilibrium)	1.500	0.777	✓

Table 2: Relaxation of a proton ensemble in a 1 keV proton background ($n_b = 10^{23} \text{ m}^{-3}$) to the Maxwellian, evolved for 8 relaxation times. KS p -values test the final speed distribution against the Maxwell speed law at the background temperature.

The same ensemble also verifies pitch isotropisation: starting from $\xi_0 = 1$, the relaxed ensemble gives $\langle \xi \rangle = 0.009$ (target 0) and $\langle \xi^2 \rangle = 0.330$ (target $1/3$), consistent with convergence to $\xi \sim \mathcal{U}(-1, 1)$. A short-time drift check on 3.52 MeV α -particles against 10 keV electrons, over a window short enough that $K(v)$ is nearly constant (Doppler shift in speed $< 2\%$), gives $\frac{d\langle v \rangle}{dt} / K(v_0) = 0.984$ (target 1), confirming the deterministic drift term independently of the noise terms.

9.3 End-to-End Equilibration Tests (Production Tracer)

`TestMaxwellianEquilibration` and `TestPitchIsotropization` in `tests/field/test_collisions.py` exercise the full production path — `trace_particles_boozers_with_collisions`, i.e. the adaptive Dormand–Prince orbital integrator coupled to the Milstein noise step, compiled C++ coefficients, real (though simplified, `BoozersAnalytic`) magnetic geometry, and a `MaxToroidalFluxStoppingCriterion` — rather than the idealised velocity-space-only integration of Section 9.2.

Test	Measured	Target	Pass
Hot start ($E_0 = 4.5 T_b$), $\langle E \rangle / T_b$	1.392	1.5	✓
Cold start ($E_0 = 0.45 T_b$), $\langle E \rangle / T_b$	1.601	1.5	✓
Pitch isotropisation, $\langle \xi \rangle$	−0.019	0	✓
Pitch isotropisation, $\langle \xi^2 \rangle$	0.326	$1/3$	✓
Pitch broadening ($\xi_0 = 0$), $\text{std}(\xi)$	0.418	> 0.05	✓

Table 3: End-to-end equilibration through the production tracer: 40 protons on a 1 keV proton background, $n_b = 10^{21} \text{ m}^{-3}$, 8 relaxation times, ITER-scale confining field. All particles remain confined and finite throughout (no losses, no NaN states).

The pitch-decay benchmark and its correction. A quantitative pitch-decay test against the naive expectation $\langle \xi(t) \rangle = \xi_0 e^{-\nu_D t}$ (the usual ASCOT5-style benchmark, which assumes v constant) failed the first time this test suite was run on Perlmutter: for 3.52 MeV α -particles

on a single deuterium background, the drag rate is $(m_a/m_b)\nu_D = 2\nu_D$, so the ensemble slows almost completely over the same window in which pitch decay is measured, and $\nu_D(v) \propto v^{-3}$ diverges as the particles thermalise — the ensemble fully isotropises ($\langle \xi \rangle \rightarrow 0$) rather than decaying exponentially at the fixed initial rate. The test was corrected to check the classical relation obtained by eliminating time between the drag and pitch equations,

$$\frac{\langle \xi(t_2) \rangle}{\langle \xi(t_1) \rangle} = \left(\frac{v(t_2)}{v(t_1)} \right)^{m_b/m_a}, \quad (35)$$

evaluated self-consistently against the simulation’s own mean speed (referencing the first saved snapshot rather than $t = 0$ cancels a geometric offset from sampling ξ along field-line-following orbits). A second, independent effect was found in the same debugging pass: in a toroidal (non-uniform- $|B|$) field the trapped-passing boundary can sit just below the injected pitch, so scattered particles fall into the trapped population where the orbit-averaged ξ is systematically pulled toward zero, roughly doubling the apparent decay rate relative to the uniform-plasma law — real neoclassical physics, but not what a velocity-space-operator test should measure. The test therefore uses a near-uniform- $|B|$ configuration ($\bar{\eta} \rightarrow 0$: no mirror force, no trapped population), giving

$$\frac{\langle \xi_2 \rangle / \langle \xi_1 \rangle}{(v_2/v_1)^{m_b/m_a}} = 1.038 \quad (\text{target } 1, \text{ tolerance } \pm 15\%),$$

with $v_2/v_1 = 0.498$ over the measurement window.

9.4 Slow Physics Suite (MPI, HPC)

Five tests marked `@pytest.mark.slow` run longer physical integrations ($N \sim 20$ –128 particles, 5–100 μs of physical time) that are impractical to run serially: an electron-drag signal-to-noise test (chosen specifically because ion-background drag is buried in noise for 3.52 MeV α -particles, whereas electron-background drag is enhanced by m_α/m_e), a cold-plasma energy-loss test, a two-background superposition test, and the pitch-broadening/decay tests of Section 9.3. The tracer’s MPI support (particles distributed across ranks, results all-gathered identically on every rank, per-particle seeds rank-independent) lets this suite run on `tests/perlmutter/run_slow_tests.sh`: all five tests pass in 77 s under 64 MPI ranks on Perlmutter, versus 3h45m running the same suite serially.

9.5 Cross-Code Validation Against ASCOT5: Uniform Plasma

`examples/collision_validation_ascot5/` mirrors the alpha slowing-down branch of ASCOT5’s own built-in Coulomb-collision physics test: N fusion alphas (birth energy 3.52 MeV, isotropic pitch, shared random seed) slow down in a uniform 1 keV hydrogen plasma. The two codes use different magnetic geometries (ASCOT5’s analytical ITER-like circular tokamak; firm3d’s BoozerAnalytic at the same R_0, B_0 scale), but since the plasma is uniform the compared quantities — mean slowing-down time to 50 keV and $\langle E(t) \rangle$ — are geometry-insensitive, isolating the collision operator itself.

9.6 Reactor-Scale Demonstration: Wistell-A / ARIES-CS

`examples/fusion_distribution_collisional/` traces 1024 fusion-born alphas (birth positions weighted by the fusion reactivity profile, isotropic pitch) for 150 ms in the Wistell-A configura-

	firm3d	ASCOT5	Agreement
Mean slowing-down time [ms]	24.915 ± 0.027	24.885 ± 0.029	+0.1%, 0.7 σ

Table 4: $N = 1024$ alphas, $n = 10^{20} \text{ m}^{-3}$ (the published ASCOT5 test-point density), $T = 1 \text{ keV}$. Also: $\max |\langle E(t) \rangle_{\text{firm3d}} - \langle E(t) \rangle_{\text{ASCOT5}}|/E_0 = 0.4\%$ over the full slowing-down window.

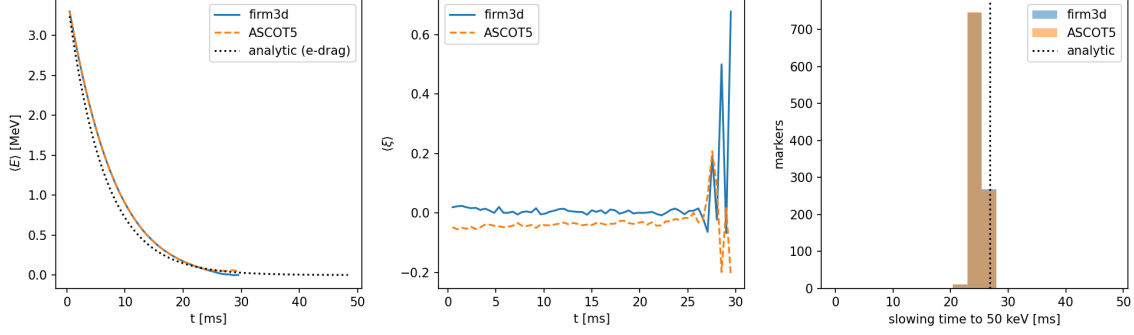


Figure 1: firm3d vs. ASCOT5, uniform 1 keV plasma, $N = 1024$: mean energy $\langle E(t) \rangle$ against the analytic electron-drag estimate $v = v_0 e^{-t/\tau_{se}}$ (left); mean pitch $\langle \xi(t) \rangle$, shown for qualitative comparison only since it is geometry-sensitive (centre); and the distribution of per-particle slowing-down times to 50 keV against the analytic Spitzer estimate (right).

tion scaled to ARIES-CS size and field strength, against a D–T–electron background with the profile shapes of Bader et al. [14] (core $n_D = n_T = 10^{20} \text{ m}^{-3}$, $T_0 = 11.5 \text{ keV}$). The identical birth ensemble is also traced collisionlessly, isolating prompt orbit losses from collisional (pitch-angle-scattering-driven) transport.

	With collisions	Collisionless
Particle loss fraction (150 ms)	19–21%	1.2%
Energy loss fraction (wall loading)	7.6%	—

Table 5: firm3d, Wistell-A/ARIES-CS scale, $N = 1024$, 1 CPU node $\times$ 128 MPI ranks, 13 min (debug queue). Particle and energy loss fractions differ because late-time collisional losses are increasingly thermalised ash carrying little energy.

The time-integrated (i.e. ergodically equivalent to a steady-state source) energy spectrum of confined alphas matches the classical slowing-down distribution $f(E) \propto \sqrt{E}/(E^{3/2} + E_c^{3/2})$ (critical energy $E_c = 266 \text{ keV}$) to a mean [relative residual] of 4.7% over 0.9–3.3 MeV (Figure 2, right).

As a literature consistency check (not a marker-level validation), Bader et al. [14] ran the same Wistell-A/ARIES-CS-scale configuration with the ANTS code and report a collisional energy-loss fraction of $\approx 8\%$ (their Fig. 3, plateau near 100 ms), with Wistell-A performing on par with W7-X. The firm3d result above (7.6% at 150 ms) is consistent, though several inputs differ (Bader et al. use $n_{e,0} = 4.8 \times 10^{20} \text{ m}^{-3}$ with $Z_{\text{eff}} = 1.13$ versus $n_{e,0} = 2 \times 10^{20} \text{ m}^{-3}$, $Z_{\text{eff}} = 1$ here, and follow particles to full thermalisation).

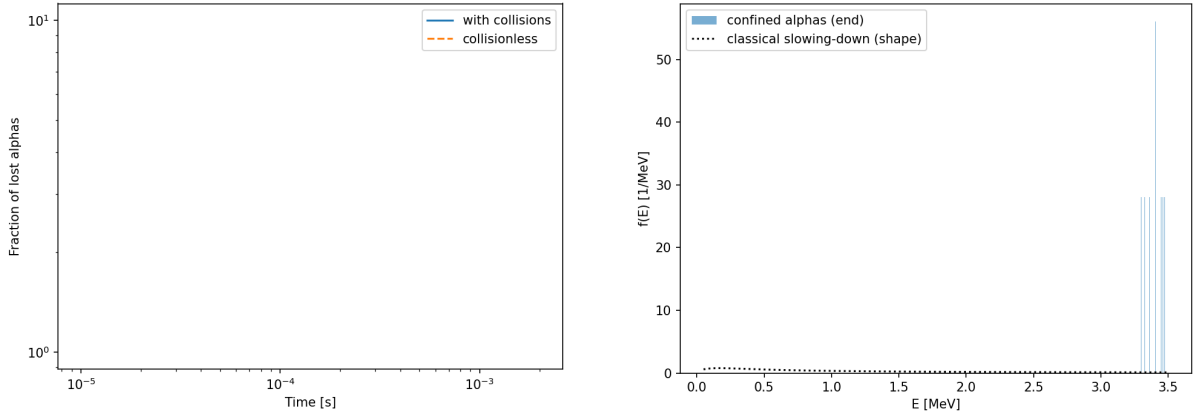


Figure 2: Left: loss fraction vs. time, with and without collisions. Right: time-integrated confined-alpha energy spectrum vs. the classical slowing-down distribution.

9.7 Direct Cross-Code Comparison in the Wistell-A Geometry

Beyond the uniform-plasma validation of Section 9.5, `examples/fusion_distribution_collisional/run_ascot5_wistell.py` builds an ASCOT5 input for the identical Wistell-A equilibrium directly from the VMEC `wout` file (a hand-padded real-space (R, ϕ, z) spline, as opposed to firm3d’s Boozer-coordinate spline of the same `booz_xform`-transformed field), with the same D–T–electron profiles, birth-energy and pitch distribution, and integration time as Section 9.6.¹

Collisional	firm3d	ASCOT5	Agreement
Particle loss fraction	19.4%	22.6%	1.7σ
Energy loss fraction	7.6%	13.4%	large discrepancy
Collisionless	firm3d	ASCOT5	Agreement
Final loss fraction	1.2%	27.2%	18.2σ

Table 6: $N = 1024$, same Wistell-A/ARIES-CS equilibrium, same birth distribution shape and profiles. Collisionless: particles retain full birth energy until lost, so particle-loss fraction is energy-loss fraction.

The 18.2σ collisionless discrepancy is noteworthy because both codes trace the *same* VMEC equilibrium (Section 9.7), so a large gap in a real physical effect (secular guiding-centre orbit diffusion from broken exact quasisymmetry) more likely reflects a numerical or methodological difference between the two field representations than two genuinely different physical predictions. Candidate explanations — none yet confirmed — include differing resolution of the fine, non-quasisymmetric ripple harmonics that drive this diffusion in each code’s respective spline, boundary-handling differences for orbits that repeatedly graze the loss boundary, and differences in the guiding-centre equations or order retained by each code. This is flagged as an open item for a future, better-suited cross-code comparison rather than a validated result.

¹Two ASCOT5 source patches were required and are documented in `run_ascot5_wistell.py`’s module docstring: a compile-time Wiener process array too small for this problem, and a near-magnetic-axis floating-point edge case in the generic ρ evaluator that a sibling function in the same codebase already guards against.

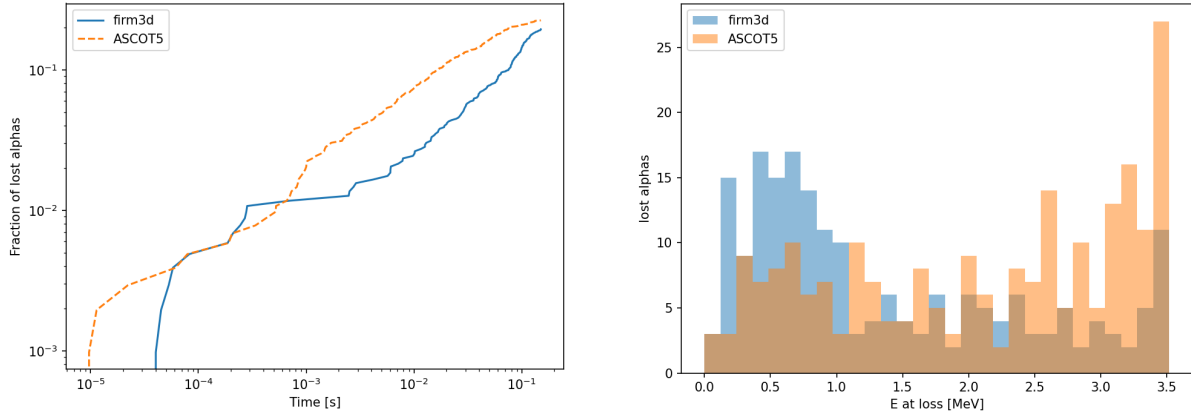


Figure 3: Collisional Wistell-A comparison. Left: loss fraction vs. time. Right: energy distribution of lost alphas – ASCOT5’s lost population skews markedly toward near-birth-energy (“prompt”) losses relative to firm3d’s, which skews toward thermalised, low-energy losses; this drives the energy-loss-fraction discrepancy despite the particle counts agreeing to 1.7σ .

10 Validity and Limitations

- **Vacuum field:** electric fields, $E \times B$ drifts, and electromagnetic perturbations are not included in the collision model (though perturbations can be combined via the existing SAW tracing infrastructure).
- **Maxwellian backgrounds:** the collision operator assumes each background species has a local Maxwellian distribution function. Distorted distributions (e.g. beam-driven) are not supported.
- **Test-particle limit:** the EP density is assumed negligibly small compared to the background, so no back-reaction on the background is included.
- **Guiding-centre approximation:** valid when the Larmor radius $r_L = mv_{\perp}/(|q|B)$ is small compared to the scale lengths of B and the density/temperature profiles. For 3.52 MeV α -particles in a 5 T stellarator $r_L \sim 5$ cm, comparable to device scale, so finite-Larmor-radius corrections may be non-negligible.
- **Zeroth-order GC collision operator:** the implementation follows Hirvijoki et al. [1] at zeroth order in ρ_*/L (Larmor radius over gradient length). The spatial diffusion term $D_X \propto D_{\perp}/\Omega_g^2$ present in the full operator [2] is neglected.
- **Slow tests:** `@pytest.mark.slow` — five physics-validation tests require $N \sim 20$ particles integrated for 5–100 μ s and are intended for HPC resources (Perlmutter or similar).

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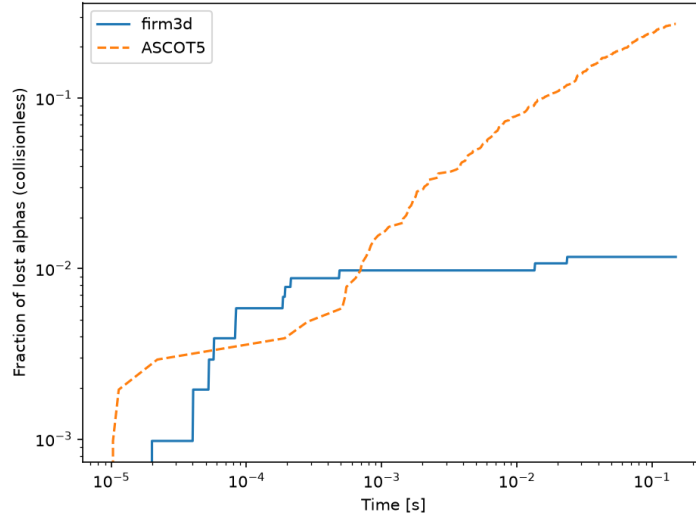


Figure 4: Collisionless (orbit-following only) loss fraction vs. time. The two codes agree at early times (~ 0.3 – 1% within the first millisecond, consistent prompt-loss physics) but diverge entirely in a late-time, slow “stochastic” loss channel with zero collisions: firm3d’s curve is flat after ~ 1 ms, while ASCOT5’s continues climbing through the full 150 ms window.

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